MIL-T-80112C
11 September 1984
SUPERSEDING
MIL-T-80112B
27 April 1979

MILITARY SPECIFICATION

THREAD AND FORM ROLLING MACHINES, ROTARY DIE, POWER DRIVEN, HORIZONTAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of electric motor driven, two die, in-feed, thru-feed, thread and form rolling machine with equipment and accessories specified herein.

1.2 Classification. This specification covers machines of the following sizes. The size required for a specific procurement shall be as specified (see 6.2.1).

Size	Rolling Pressure
1	25,000 lbs.
2	30,000 lbs.
3	45,000 lbs.
4	70,000 lbs.
5	200,000 lbs.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-TES, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3449

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/IEEE 268 - Metric Practice.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standards for.

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, (see 6.2.1) a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of producing threads or forms in workpieces of solid stock or hollow stock in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed,

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remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Environmental protection. The machine shall be designed and constructed such that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at the point of the installation.

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or the machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

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3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than .001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for their application. Gears shall be applied in a manner to preclude either premature damage, wear or failure. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 Components. The thread and form rolling machine shall consist essentially of a base, die spindles, drive system, size adjusting mechanism, coolant system, hydraulic system, electrical system and such other components as are required to perform as specified herein.

3.4.1 Base. The base shall be a casting or weldment, sufficient in strength and rigidity to support the components mounted thereon and to withstand all forces encountered when the machine is operating at its full rated capacity. The base shall have an access door or removable panel to permit servicing of any internal components.

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3.4.2 Die spindles and drive. The roll die spindles shall be sufficient in size and strength to withstand the rolling forces imposed upon them. The spindles shall be mounted into the spindle housings on anti-friction bearings. The spindles shall be powered by an electric motor through a gear box and suitable drive transmission system.

3.4.3 Size adjusting mechanism. The machine shall have a size adjusting mechanism graduated in increments of 0.001 inch to provide final sizing of the thread or form.

3.4.4 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a baffle or a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.5 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.6 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4.6.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

3.4.6.2 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar ageing devices is permitted only for voltage surge protection to other solid state components.

3.4.6.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

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TABLE I. Sizes and capacities.

CHARACTERISTICS	Size 1	Size 2	Size 3	Size 4	Size 5
Rolling Pressure (lbs)	25,000	30,000	45,000	70,000	200,000
Pitch Range (TPI) Unified	8-80	5-80	5-80	5-80	4-80
Spindle Speed (RPM) (Variable to at least)	500	250	250	230	475
In-feed Rolling:					
Diameter, Inches	2	2-1/2	3	4	6
Pitch (TPI) Unified	8	5	5	5	4
Pitch (TPI) Acme	10	6	6	4	2
Pitch (DP) Worm	16	20	9	6	4
Thread Length, Inches	2	3	5	5-3/4	9-3/4
Thru-feed Rolling:					
Diameter, Inches	2	2	2	4	4
Pitch (TPI) Unified	8	5	8	8	4
Length, Feet	20	20	20	20	20

Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure, or less, to the larger figure, or more.

3.6 Performance. The machine shall be capable of rolling right or left hand external threads within its capacity, as listed in table I, and special forms such as splines and knurls. The machine shall be capable of in-feed and thru-feed rolling. The machine shall provide for manual, semi-automatic and automatic die cycling for in-feed rolling operations and automatic feeding for thru-feed operations. The capacity of the machine shown in table I is based on its maximum capability in mild steel rods having 229 Brinell or 20.5 RC hardness.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

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3.7 Alignment tolerances. Each machine shall meet the alignment accuracies of table II.

TABLE II. Alignment accuracies.

Test	Permissible Error
Spindle runout	0.002 inch TIR
Spindle endplay	0.001 inch TIR
Spindle axial alignment	0.002 inch

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Marking of instruction plates, indicating plates and name plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

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Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Elapsed time meter. Unless otherwise specified (see 6.2.1) the machine shall be equipped with an elapsed time meter. The elapsed time meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching the maximum accumulative hours, the meter readout shall automatically revert to zero and continue to totalize time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.3. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.10.

4.6 TESTS

4.6.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Alignment accuracy tests. Each machine shall be tested to determine that it complies with the alignment accuracy requirements of table II (see 3.7).

4.6.3 Performance tests. Each machine shall be operated to thread ten workpieces by the in-feed method and five workpieces, not less than 18 inches long, by the thru-feed method. Die pattern and workpiece diameter shall be that which is the largest capacity in Table I for the machine being tested. Thread form and tolerances shall conform to FED-STD-H28.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D 3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for production rolling of threads and forms on solid and heavy walled hollow stock.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Size required (see 1.2).

c. When first article is required for inspection and approval (See 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.4).

h. Painting, if different (see 3.3.5).

i. Electrical system and input power supply, if different (see 3.4.6).

j. Motor enclosure, if different (see 3.4.6.1).

k. Size and capacity, if different (see 3.5).

l. Additional performance features, if required, (see 3.6.1).

m. Specify and fully describe optional equipment required if different (see 3.8).

n. Specify applicable repair parts, if required (see 3.9).

o. Lubrication chart or plate, if different (see 3.10.1).

p. Nameplate, if different (see 3.10.2).

q. Elapsed time meter, if required (see 3.11).

r. First article inspection, if different (see 4.3).

s. Quality conformance inspection, if different (see 4.4).

t. Level of packaging, packing and marking, if different (See 5.1).

6.2.2 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List and shall be furnished in accordance with each specified DD Form 1664, Data Item Description as shown in the DD Form 1423.

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on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacturer's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activity:
Army - AL
Air Force - 84
DLA- GS

User Activities:
Navy - MC

Preparing Activity:
DLA-IP

Project Number:
3449-0014

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-80112C	Thread and Form Rolling Machines, Rotary Die, Power Driven, Horizontal

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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